

2.6.

KIRPICILAR.

1/23



$\Rightarrow V_A - V_K \geq 0$ ise Diyot iletimde } ideal diyot için.
 $V_A - V_K < 0$ ise Diyot kesimde

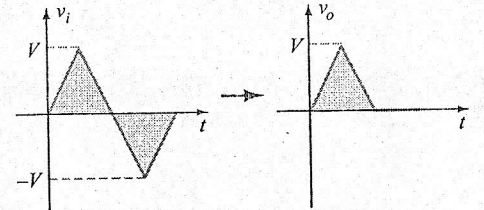
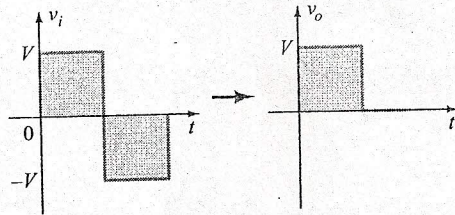
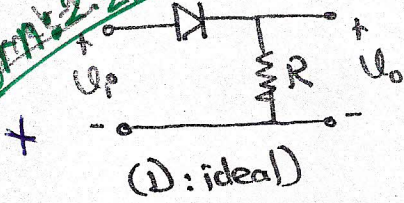
$V_A - V_K \geq V_D$ ise Diyot iletimde } ideal olmayan diyot için.
 $V_A - V_K < V_D$ ise Diyot kesimde.

$$(V_D = 0.7V \text{ Si} \\ = 0.3V \text{ Ge})$$

2.6.1

— Diyodların seri kolle olması:

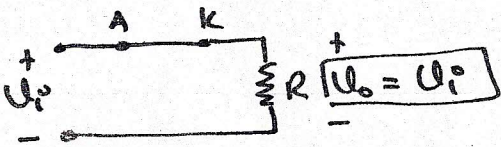
ÖRNEK-2.21



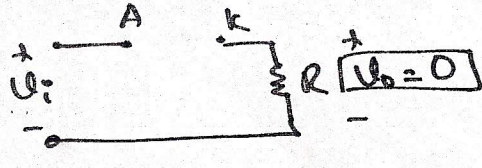
$V_A = U_i$
 $V_K = 0V$ } $V_A - V_K = U_i - 0 = U_i \Rightarrow U_i \geq 0$ ise Diyot iletimde $\Rightarrow$ 1.azona.
 $U_i < 0$ ise Diyot kesimde.

2.azona:

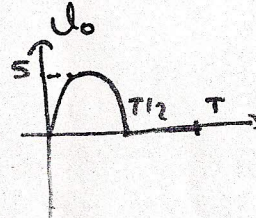
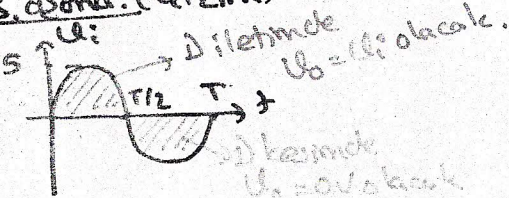
D iletimde iken ($U_i \geq 0$ iken):



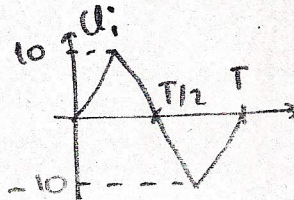
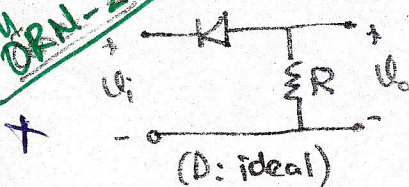
D kesimde iken ($U_i < 0$ iken)



3.azona: (Gizim)



ÖRNEK-2.22



olun U_o 'i çiziniz.

1.azona:

$$V_A = 0V.$$

$$V_K = U_i$$

$V_A - V_K \geq 0$ ise

yani

$U_i \geq 0$ ise Diyot iletimde.

$U_i \leq 0$ ise Diyot kesimde.

$U_i > 0$ ise Diyot kesimde.

2.azona

D iletimde ise;

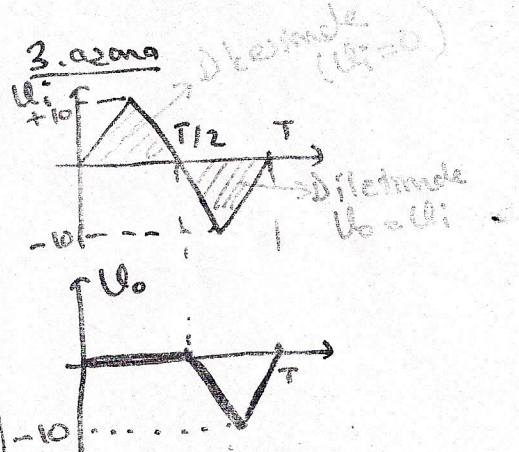


$$U_o = U_i$$

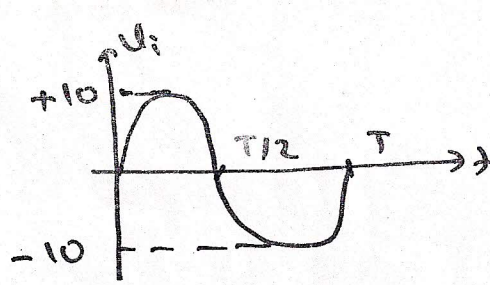
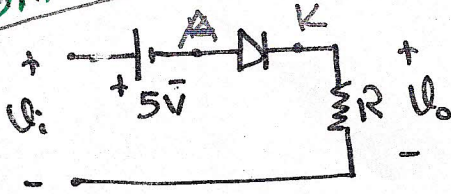
D kesimde ise;



3.azona



ÖRNEK 2.23 ✓



ise $U_o = ?$

çiziniz.

(Not: Diyod ideal)
değil ve 5i

Gözüm:

1. Aşama:

$V_A = ?$

Gevre yazılabilir.

$$-U_i + 5 + V_A = 0$$

$$V_A = U_i - 5$$

veya ;

V_A ile U_i aynı, 5V

zıt polaritede ve $\Rightarrow V_A = U_i - 5$

$$V_K = 0V$$

$$\Rightarrow V_A - V_K \geq 0.7 \text{ ise iletim}$$

$$< 0.7 \text{ ise kesim.}$$

$$\hookrightarrow U_i - 5 \geq 0.7 \Rightarrow \boxed{U_i \geq 5.7 \text{ ise Diletimde}}$$

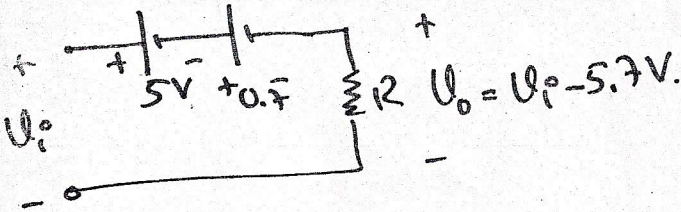
$$U_i < 5.7 \text{ ise D kesimde}$$

Burada $\uparrow$

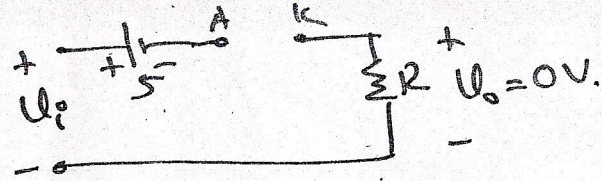
$V_K = 0$ old. için

2. Aşama:

D iletimde iken:

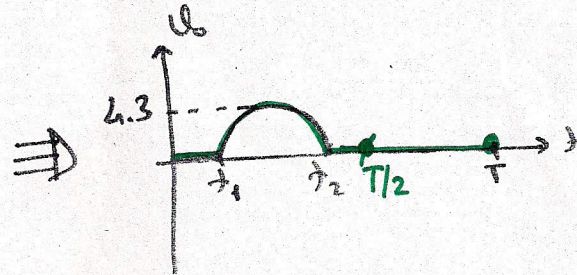
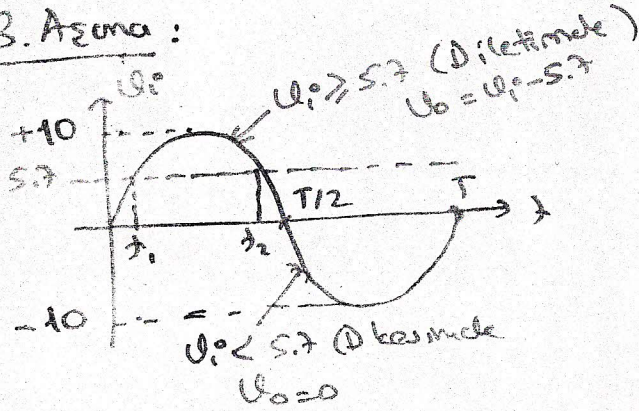


D kesimde iken:



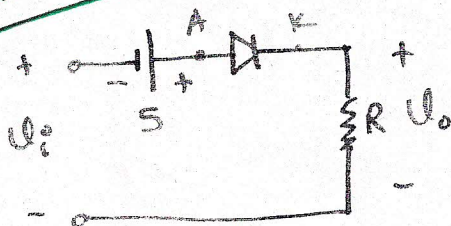
b) Uager çizis

3. Aşama:



(Not: bir iki sinyal perçacığını - ve + ile kaydırarak çizilebilir. $\uparrow$ gibi.)

ÖRNEK 2.24 ✓



Aynı U_i .

$$V_A = U_i + 5$$

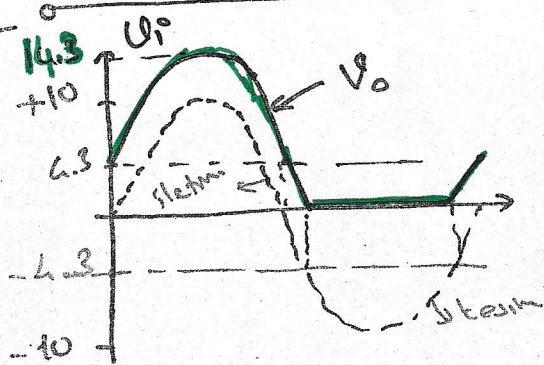
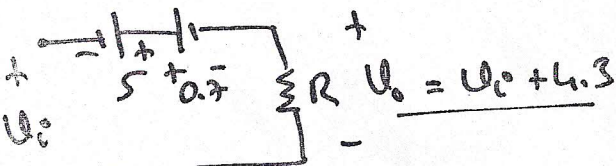
$$V_K = 0$$

$$U_i + 5 \geq 0.7 \Rightarrow U_i \geq -4.3 \text{ ise iletim.}$$

$$U_i < -4.3 \text{ ise kesim.}$$

$\hookrightarrow$

İletimde:

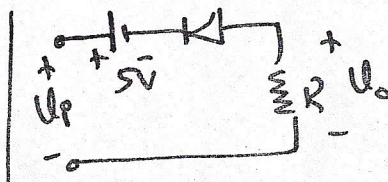


kesimde

$$U_o = 0V$$

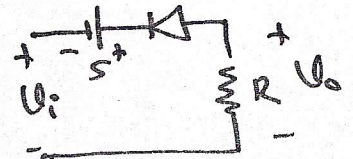
Ödev:

* *



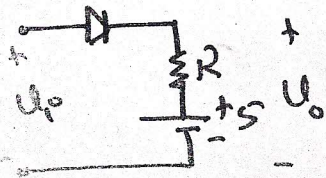
(a)

U_i : aynı.



(b)

ÖRNEK-2.25 ✓



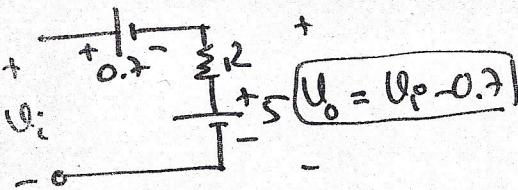
$$\left. \begin{array}{l} V_A = U_i \\ V_K = 5V \end{array} \right\}$$

$U_i - 5 > 0.7$ ise D iletimde.

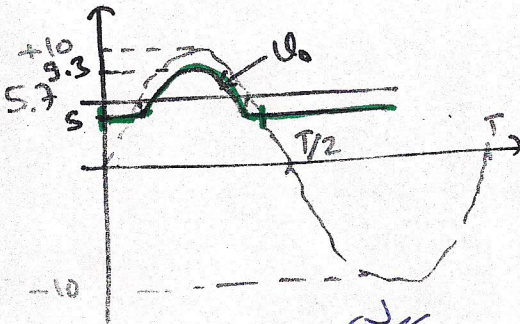
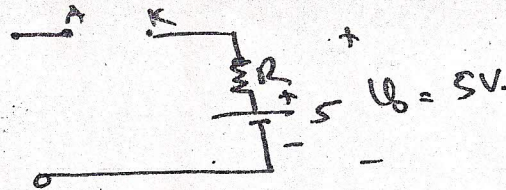
$U_i > 5.7$ ise D iletimde

$U_i < 5.7$ ise D kesimde

D iletimde iter:

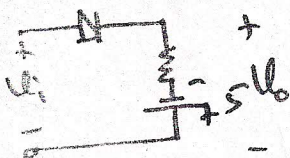


D kesimde iter:



ÖRNEK-2.26

ÖDEV



$$\left. \begin{array}{l} V_A = U_i \\ V_K = -5V \end{array} \right\}$$

$U_i - (-5) > 0.7$

$U_i > -4.3$ De iletim

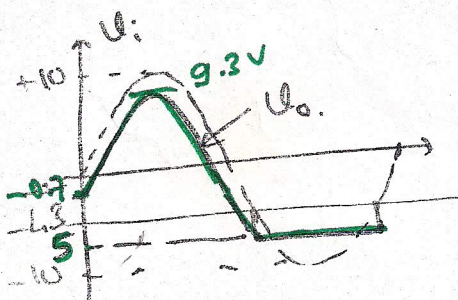
$U_i < -4.3$ De kesim

İletimde:

$$U_o = U_i - 0.7$$

kesimde:

$$U_o = -5V$$



Ödev:

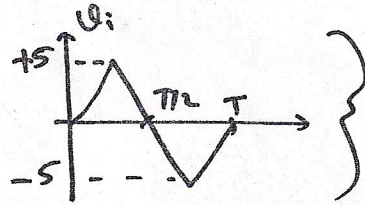
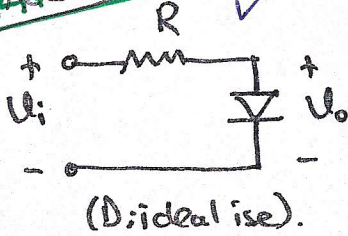
* *

Örnek 2.25 ve 2.26'dıyı Diyodlar test iter gösteririz.

2.6.2 Diyalları Paralel Kolda Olması:

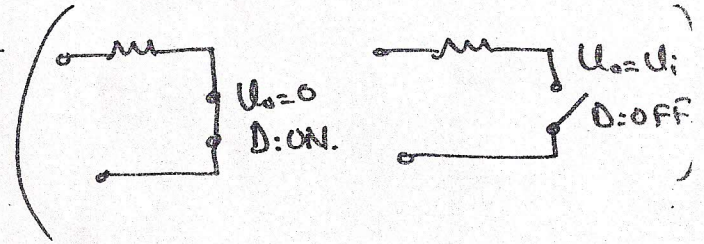
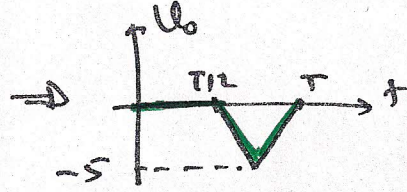
26

ÖRNEK-2.27

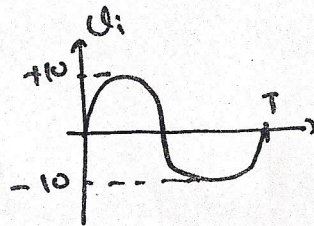
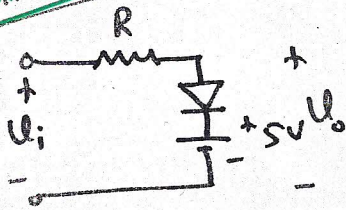


$$\left. \begin{array}{l} V_A = U_i \\ V_K = 0 \end{array} \right\} \begin{array}{l} U_i \geq 0 \text{ ise } D \text{ iletken} \\ U_i < 0 \text{ ise } D \text{ kesme} \end{array}$$

D iletken iken $\Rightarrow U_o = 0V$.
D kesme iken $\Rightarrow U_o = U_i$



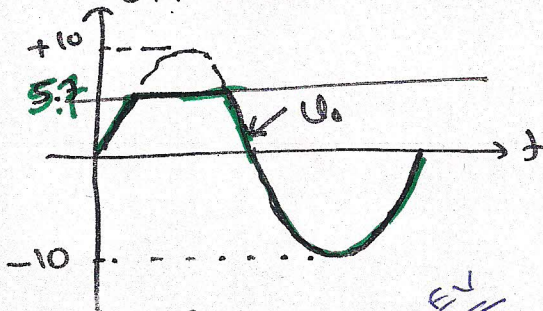
ÖRNEK-2.28



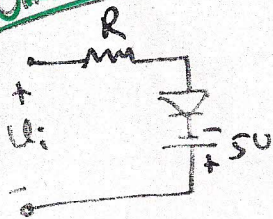
D: Si (ideal değil).

U_o 'i çiziniz.

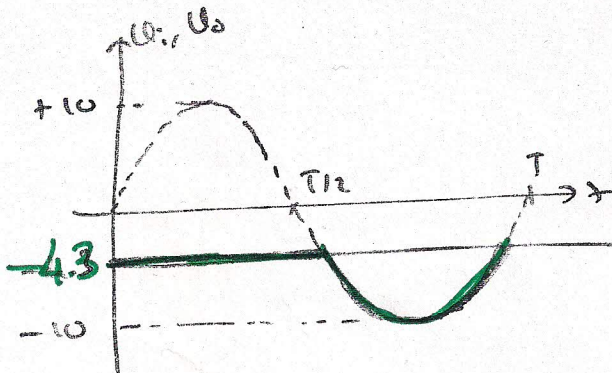
$$\left. \begin{array}{l} V_A = U_i \\ V_K = 5 \end{array} \right\} \begin{array}{l} U_i - 5 \geq 0.7 \Rightarrow U_i \geq 5.7 \text{ ise } D \text{ iletken} \Rightarrow U_o = 5.7V \\ U_i < 5.7 \text{ ise } D \text{ kesme} \Rightarrow U_o = U_i \end{array}$$



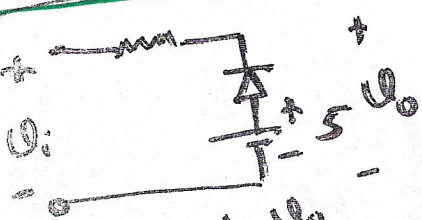
ÖRNEK-2.29



$$\left. \begin{array}{l} V_A = U_i \\ V_K = -5 \end{array} \right\} \begin{array}{l} U_i - (-5) \geq 0.7 \text{ ise } D \text{ iletken} \\ U_i \geq -4.3 \text{ ise } D \text{ iletken} \Rightarrow U_o = -4.3 \\ U_i < -4.3 \text{ ise } D \text{ kesme} \Rightarrow U_o = U_i \end{array}$$

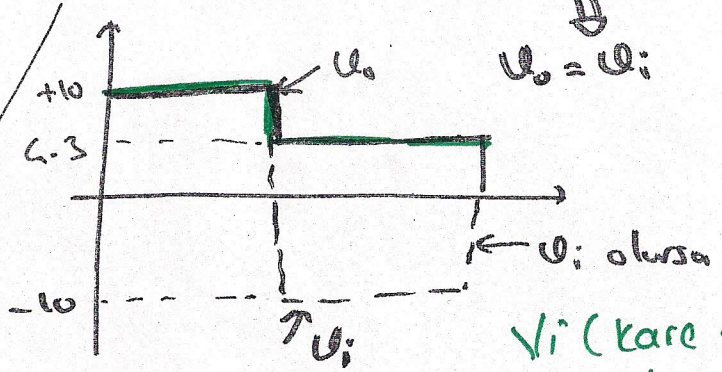
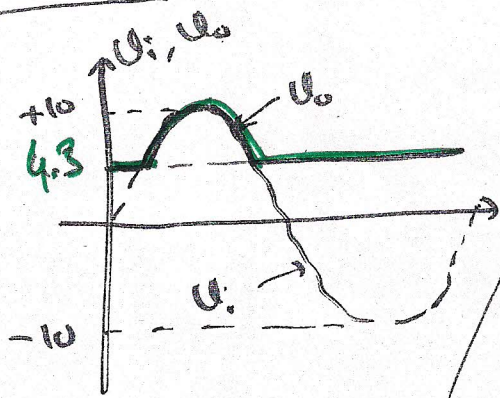


ÖRNEK-2.30 ✓



$$\begin{aligned} V_A &= 5V \\ V_K &= U_i \end{aligned}$$

$5 - U_i > 0.7 \rightarrow U_i < 4.3$ ise iletkir
 $\Downarrow$
 $U_o = 4.3V$
 $U_i > 4.3$ ise kesirli
 $\Downarrow$
 $U_o = U_i$

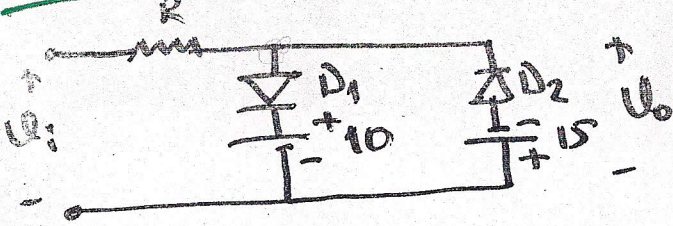


U_i (kare dalga olursa)

Örnek:

Örnek 2.30: -5V iten **

ÖRNEK-2.31 ✓



$U_i : \pm 20V (V_p)$
 Sinüs.

D: ideal. $U_o = ?$

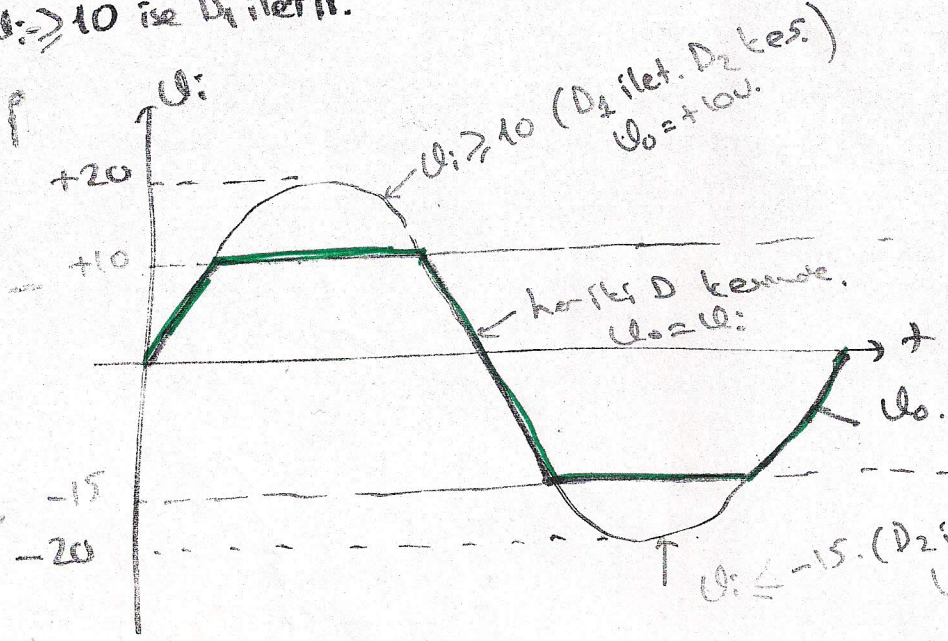
D1:

$U_i - 10 > 0$
 $U_i > 10$ ise D_1 iletkir.

D2:

$-15 - U_i > 0$
 $U_i < -15$ ise D_2 iletkir.

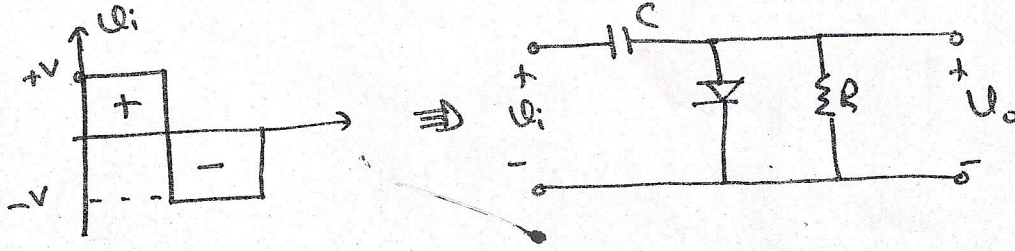
$10 > U_i > -15$ iten
 D_1, D_2 : kesirli.



27. KENETLEVICİLER.

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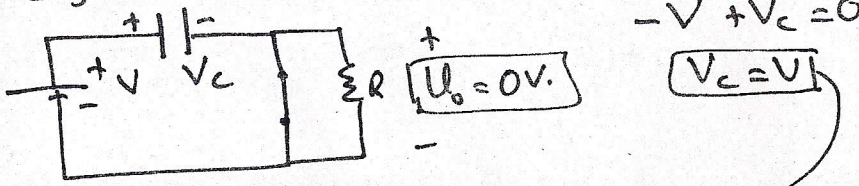
Kenetleme devreleri, bir sinyalin DC seviyesini farklı bir DC düzeye kenetleyen devrelerdir. R, C ve diyot kullanılır.



- + C - şarj - akım (+)'den girer
- * C - deşarj - akım (-)'den girer
- * Diyot ↓ DC seviye ↓
- * Diyot ↑ DC seviye ↑

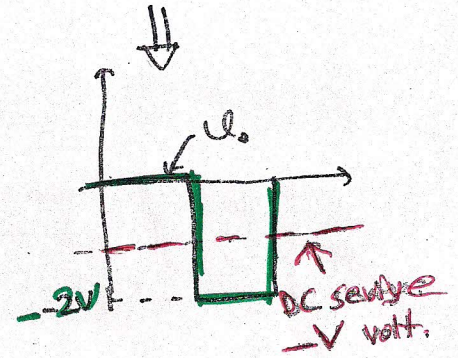
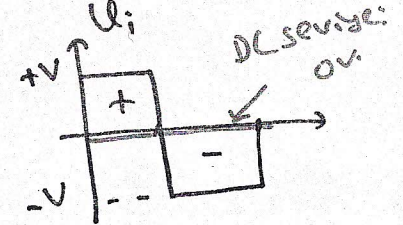
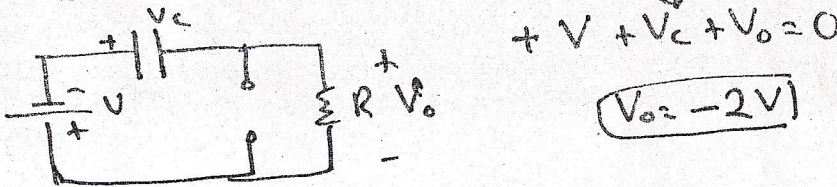
(+) alternansta:

D geçirir ve C, şarj olur.

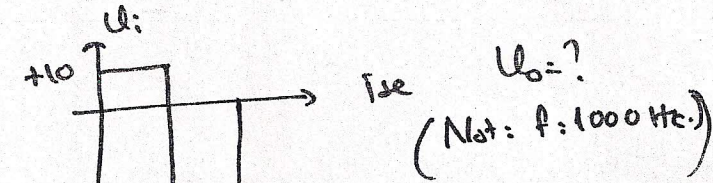
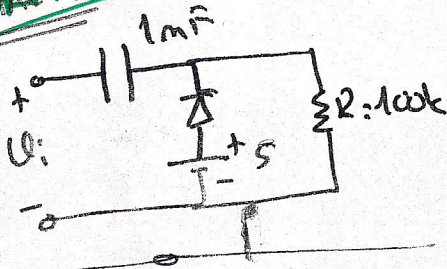


(-) alternansta:

D keşirir ve C, R üzerinden deşarj olur.

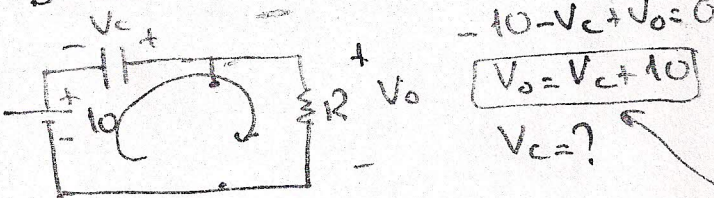


ÖRNEK-2.32



(+) alternansta

D keşirir ve C şarj olur.



$$-10 - V_C + V_o = 0$$

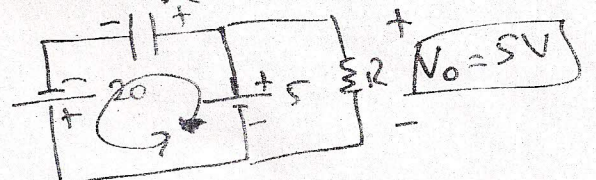
$$V_o = V_C + 10$$

$$V_C = ?$$

$$V_o = 35V$$

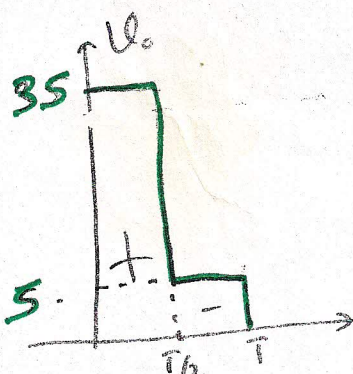
(-) alternansta

D iletimde. C şarj olur.



$$-20 - 5 + V_C = 0$$

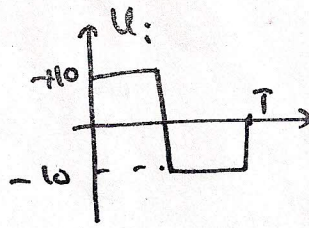
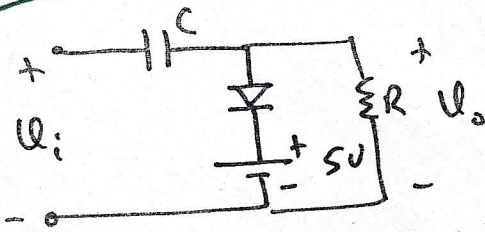
$$V_C = 25V$$



U_i R'n DC seviyesi ⇒ -5V idi;

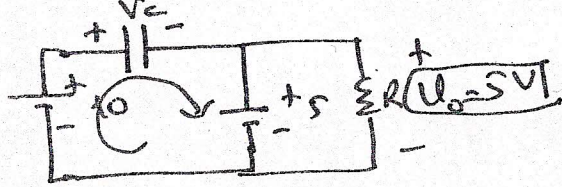
U_o R'n DC seviyesi ⇒ 20V oldu.

ÖRNEK-2.33



(+) alternansta D iletimde

$C \Rightarrow \text{kısa}$.

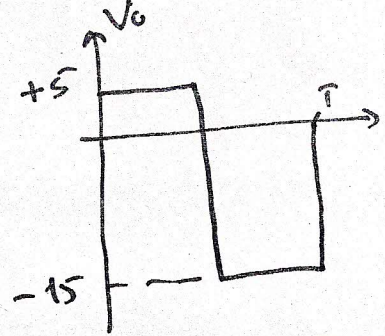


Not: C kısa olurken akım (+)'da girer.
değerli olurken (-)'de girer.

buzetilde V_c polaritesi belirlenir.

$$-10 + V_c + 5 = 0$$

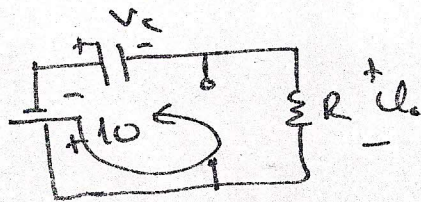
$$V_c = 5V$$



DC seviye 0'dan ~~5~~ V 'a düşer.

(-) alternansta D kesimde

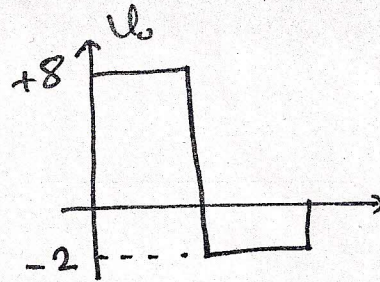
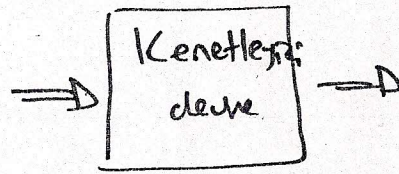
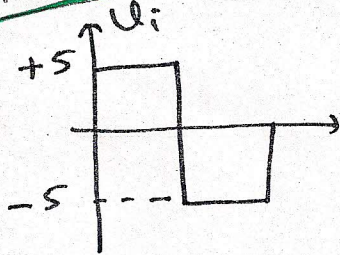
$C \Rightarrow \text{değerli}$.



$$-10 - V_o - V_c = 0$$

$$V_o = -15V$$

ÖRNEK-2.34

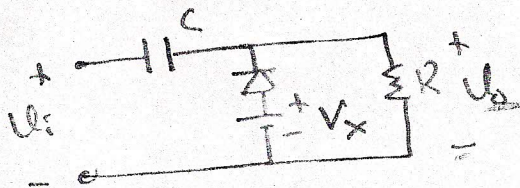


TASARLAYIN.

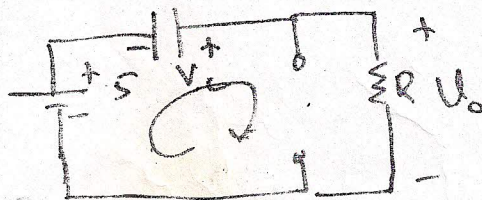
Çöz:

D seçimiye bakarsa DC seviye azalır, yukarıya bakarsa artar. Bu nedenle:

V_x : bilinmeyen bir DC voltajı.



(+) alternansta D kesimde, $C \Rightarrow \text{değerli}$.



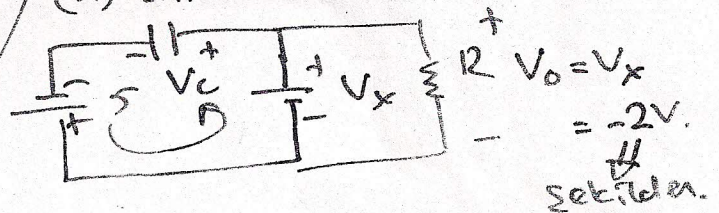
$$-5 - V_c + V_o = 0$$

$$V_o = 8V \text{ (sekiiden)}$$

$$-5 - V_c + 8 = 0$$

$$V_c = 3V$$

(-) alternansta D ilel. $C \Rightarrow \text{kısa}$.



ve ya;

$$-5 - V_x + V_c = 0$$

$$-5 - V_x + 3 = 0$$

$$V_x = -2V$$

